

Applied Machine Learning for Implied Volatility Forecasting

Tobias Persson[†] & Anton Johanson[‡]

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Abstract

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Supervisor: Tobias Sichert, Department of Finance, Stockholm School of Economics

Examinator: Gualtiero Azzalini, Department of Finance, Stockholm School of Economics

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[†]tobias.ct.persson@outlook.com

[‡]anton13johanson@gmail.com

Table of Contents

1	Introduction	2
2	Literature Review	2
3	Data Gathering and Cleaning	2
3.1	Trading Window	2
3.2	Feature Sets	3
3.2.1	$\mathcal{M}_{\text{HAR}}$	4
3.2.2	$\mathcal{M}_{\text{ALL}}$	5
3.3	Summary Statistics	7
4	Methodology and Empirical Design	8
4.1	Realized Variance and Quarticity	8
4.2	Benchmark Models	9
4.2.1	HAR Model	9
4.2.2	LogHAR Model	10
4.2.3	LevHAR Model	10
4.2.4	SHAR Model	10
4.2.5	HARQ Model	11
4.3	Regularized Linear Models	11
4.3.1	Ridge Regression	12
4.3.2	LASSO Regression	13
4.3.3	Elastic Net	13
4.4	Tree-based Models	13
4.4.1	Bagging	13
4.4.2	Random Forest	13
4.4.3	Gradient Boosting	13
4.5	Neural Network	14
4.5.1	Feed-forward Neural Architecture	15
5	Empirical Results	17
5.1	Results from $\mathcal{M}_{\text{HAR}}$ Feature Set	17
5.2	Results from $\mathcal{M}_{\text{ALL}}$ Feature Set	18
6	Discussion	19
7	Concluding Remarks	19
8	Appendix	19

1 Introduction

2 Literature Review

The body of literature on volatility modeling in general, and more complex machine learning algorithms in particular, has expanded in recent years, as emerging technologies have made these models more accessible to the general public.

3 Data Gathering and Cleaning

The core of this paper lies in the application of using 1-minute intraday high-frequency trading data for the EURO STOXX 50 index, sourced from the Stockholm School of Economics. The initial sample spans the period from January 1998 to August 2020. However, after conducting a careful and extensive data cleaning process, we end up with a full sample covering the period from January 1999 to August 2020.

The EURO STOXX 50 index is composed of the 50 largest and most liquid companies in the Eurozone, covering about 20 different industry sectors. Therefore, the EURO STOXX 50 is often regarded as the European equivalent of the S&P500 index, making it the natural benchmark for European equity markets.

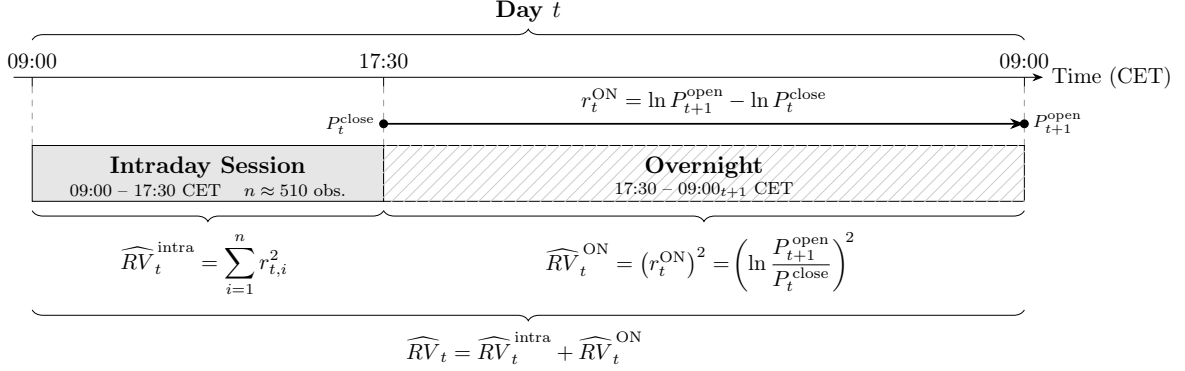
The cleaned sample spans 5,870 trading days and encompasses several distinct macroeconomic regimes, including the Dot-com Bubble (2000 - 2002), the Global Financial Crisis (GFC, 2008 - 2009), the European Sovereign Debt Crisis (Euro Crisis, 2010 - 2012), a prolonged low-volatility expansion period (2016 - 2019), and the Covid-19 market turmoil of March 2020. This broad scope of market conditions makes the EURO STOXX 50 an especially demanding and informative testing ground for volatility forecasting models, as it allows us to evaluate model performance both in tranquil periods and during episodes of profound market uncertainty.

3.1 Trading Window

The trading session for the EURO STOXX 50 runs from 09:00 to 17:30 CET, yielding approximately $n \approx 510$ 1-minute log-returns per trading day. Daily realized variance is constructed as the sum of squared intraday log-returns augmented by the squared overnight return - the return from the next days opening price at 09:00 to the current day's closing price at 17:30. This decomposition is illustrated in Figure (1).

It should be noted that the opening hours of the EURO STOXX 50 has shifted over the course of the sample period. Following a careful inspection of the intraday trading data at an aggregate level, we standardize the trading window to 09:00 - 17:30 CET for the entire sample period, ensuring consistency in the number of intraday observations used to construct daily realized variance. Furthermore, the dataset retains half-days and select public holidays on which trading has historically taken place. As these sessions

Figure 1: Trading Window and Daily Realized Variance Construction



Notes: The figure illustrates the 24-hour decomposition underlying the daily realized variance (RV) estimator for the EURO STOXX 50 index. The intraday trading session runs from 09:00 to 17:30 CET, yielding approximately $n = 510$ one-minute log-returns per trading day ($r_{t,j}$, $j = 1, \dots, n$). To obtain a complete close-to-close measure of daily variance, the squared overnight log-return - the return from the close of day t at 17:30 to the open of day $t+1$ at 09:00 - is added to the intraday sum.

involve fewer trading hours than a standard full day, the realized variance recorded on such days is expected to be lower, reflecting the reduced window over which intraday price variation can accumulate.

3.2 Feature Sets

In volatility forecasting, the selection of explanatory variables plays an important role in determining predictive performance. Albeit the traditional HAR framework relies on a small set of realized measures that capture heterogeneous persistence of volatility across different time horizons, novel machine learning techniques have allowed researchers and practitioners to incorporate additional financial and macroeconomic variables to improve forecasting performance.

To examine how the information set affects forecasting performance, this study follows the approach of Christensen et al. (2023) and constructs two different feature sets. The first feature set, denoted $\mathcal{M}_{\text{HAR}}$, contains only the core variables of the HAR framework, namely daily, weekly, and monthly trailing realized variance components. All respective models' forecasting performance are first evaluated using only this data set. This provides an apples-to-apples comparison when comparing models based on the same variables and data.

The second feature set, denoted $\mathcal{M}_{\text{ALL}}$, extends the baseline $\mathcal{M}_{\text{HAR}}$ feature set by incorporating a broader set of financial and macroeconomic predictors important in the European context to forecast volatility. By expanding the feature set, this specification allows novel machine learning models to exploit nonlinear relationships and interactions among a larger number of predictors.

Comparing the forecasting performance across these two feature sets makes it possible to evaluate whether improvements in predictive accuracy primarily stem from the use of more flexible modeling techniques or from the inclusion of additional informative

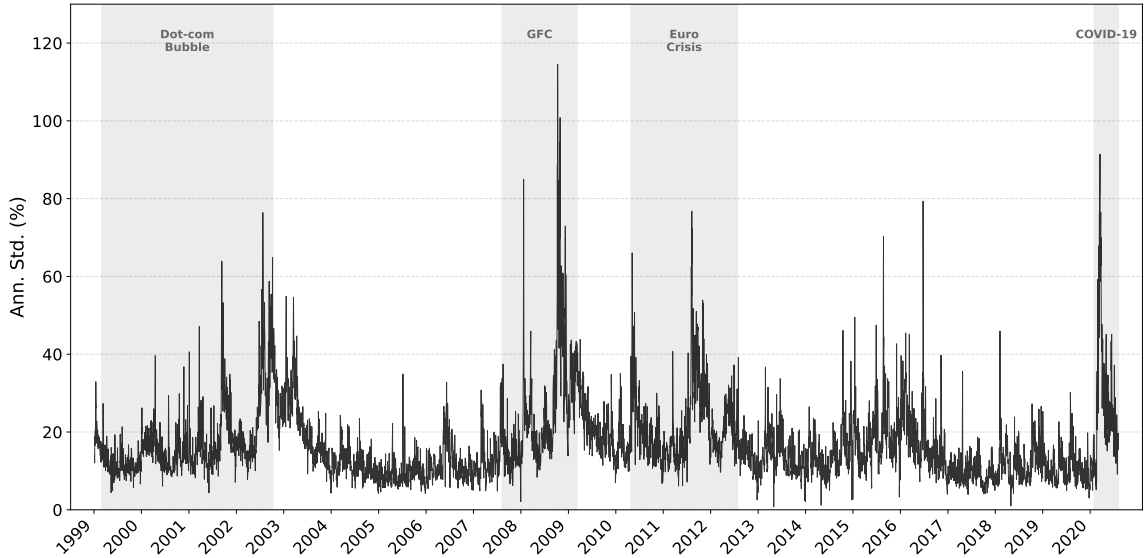
predictors.

3.2.1 $\mathcal{M}_{\text{HAR}}$

As previously noted, the $\mathcal{M}_{\text{HAR}}$ feature set is only composed of the trailing daily, weekly, and monthly realized variance compositions. The forecasting performance of all models presented in this paper are first evaluated using only this feature set. This approach will allow us to determine whether more complex and sophisticated machine learning techniques are superior compared to the conventional set of HAR models using the same predictors. However, it should be noted that the $\mathcal{M}_{\text{HAR}}$ feature set also includes variables that are specific to the various extensions of the HAR universe. For instance, the square-root of realized quarticity appears as an interaction term with realized variance in the HARQ model, but not in any of the other models.

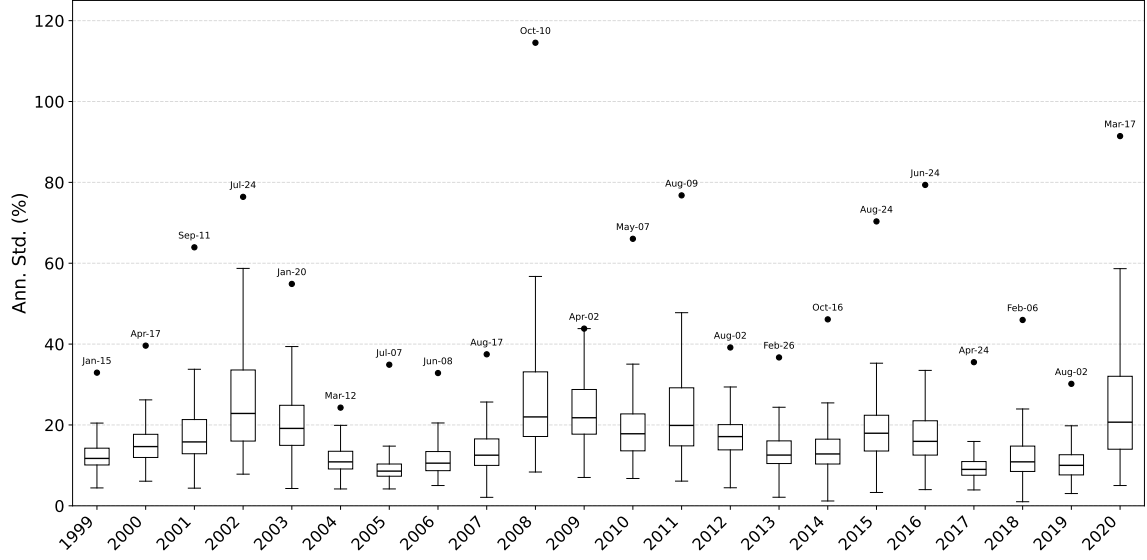
All data used to construct the $\mathcal{M}_{\text{HAR}}$ feature set are derived from the trading data set retrieved from the Stockholm School of Economics. For illustrational purposes, the distribution of annualized daily realized volatility, not variance, is depicted in Figures (2) and (3) over the entire sample period.

Figure 2: Annualized Daily Realized Volatility (1999 - 2020)



Notes: The figure plots annualized daily realized volatility for the EURO STOXX 50 index over the sample period January 1999 - August 2020. Shaded regions indicate periods of elevated market stress identified as distinct volatility regimes: the Dot-com Bubble (March 1999 – October 2002), the Global Financial Crisis (GFC, August 2007 – March 2009), the European Sovereign Debt Crisis (Euro Crisis, April 2010 – July 2012), and the COVID-19 pandemic (February 2020 – August 2020). To annualize the numbers, we assume an average of 252 trading days per year.

Figure 3: Distribution of Annualized Daily Realized Volatility (1999 - 2020)



Notes: The figure illustrates a box plot of the annual distribution of daily realized volatility for the EURO STOXX 50 over the sample period January 1999 - August 2020. Each box displays the interquartile range of daily annualized realized volatility within a given year, with the horizontal line in the box denoting the median. The dot above each box marks the single highest volatility observation of the year, with the corresponding date indicated above. Note that the box for 2020 only includes data from January till August which marks the end of our sample period. To annualize the numbers, we assume an average of 252 trading days per year.

3.2.2 $\mathcal{M}_{\text{ALL}}$

While the $\mathcal{M}_{\text{HAR}}$ feature set is restricted to realized measures derived solely from high-frequency trading data, the $\mathcal{M}_{\text{ALL}}$ feature set expands the information set by incorporating a broader range of financial and macroeconomic predictors. In contrast to $\mathcal{M}_{\text{HAR}}$, which captures the endogenous persistence of volatility through its heterogeneous autoregressive structure, $\mathcal{M}_{\text{ALL}}$ integrates exogenous sources for information that reflect forward-looking expectations, macroeconomic conditions, commodity price dynamics, and global market spillovers. According to Christensen et al. (2023), such an enriched feature set allows the models to exploit a wider set of economic signals and potential nonlinear interactions, thereby providing a more comprehensive framework for forecasting volatility.

1. *Endogenous Volatility and Risk Expectations*

The VSTOXX index, maintained by Eurex and computed by Deutsche Börse, measures the implied volatility of the EURO STOXX 50 index derived from options prices with a 30-day horizon. As a forward-looking measure of market-implied uncertainty, the VSTOXX index captures the risk premium that investors demand for bearing volatility risk, and is thus commonly used to measure the level of investor fear in European equity markets. The inclusion of the VSTOXX is motivated by the well-documented leverage effect and volatility feedback channel, whereby rising implied volatility tends to predict elevated realized volatility in subsequent periods, see Blair et al. (2001) and Bekaert and Wu (2000). Furthermore, Christensen et al. (2023) show that implied volatility subsume much of the information contained in historical realized variance for forecasting purposes. We include

both the level $VSTOXX_t$ and its logarithm $\log(VSTOXX_t)$ to accomodate the well-known right-skewness of implied volatility. Daily VSTOXX index data was obtained from the STOXX website.

2. Macroeconomic and Systemic Risk Indicators

Similarly to [Christensen et al. \(2023\)](#), we adopt the Economic Policy Uncertainty (EPU) index originally constructed by [Baker et al. \(2016\)](#). The index is constructed by counting the frequency of articles in major European newspapers that simultaneously contain terms related to the economic uncertainty and policies prevailing in the Eurozone. These counts are normalized by the total number of articles to ensure comparability over time and across sources. Note that the index is computed on a monthly basis, and we therefor compute a one month lagged version of this index to prevent look-ahead bias. The data is retrieved from the Economic Policy Uncertainty website.

In addition to the EPU index, we extend our paper by including the Composite Indicator of Systemic Stress (CISS) index developed [Hollo et al. \(2012\)](#) and provided by the European Central Bank (ECB). The daily index measures systemic risk in the European financial system by aggregating stress signals across multiple market segments, including equity, bond, money, and foreign exchange markets, as well as financial intermediaries. The CISS index is constructed from a set of market-based indicators such as volatility measures, credit spreads, liquidity indicators, which are transformed into standardized stress measures. We extract the daily CISS index from the ECB website.

3. Commodity and Inflation-Linked Factor

Contrary to [Christensen et al. \(2023\)](#), who focus on DJIA constituents in the US market, our European setting motivates the inclusion of the Brent crude oil price as an additional predictor. This decision stems from Europe’s status as a net energy importer and the significant energy sector weight in the EURO STOXX 50 index, which increases the sensitivity of equity markets to energy price fluctuations. As a key global benchmark, the Brent crude oil price captures commodity price shocks that are closely linked to inflationary pressures and macroeconomic uncertainty, see [Jawadi et al. \(2023\)](#) for further details. The daily European Brent crude oil price is obtained from the US Energy Information Administration (EIA).

4. Global Market Spillovers

To account for global spillover effects, we extend the specification of [Christensen et al. \(2023\)](#) by including both Asian and US market spillovers, motivated by the position of the European markets in the global trading day. To account for Asian market spillovers, we retrieve daily returns from the Nikkei 225 index from the Federal Reserve Economic Data (FRED) website. Note that these daily returns are predetermined relative to the opening of the European markets. All days corresponding to Japanese holidays, where there is now trading activity, we proceed to forward-fill this gap.

Alongside the Nikkei 225 index, we retrieve daily returns for the S&P500 from the Wallstreet Journal Markets website. Note that contrary to the Asian markets, daily returns from the S&P500 are not predetermined relative to the opening of the European markets. Therefore, we lag the daily returns from the S&P500 by one day. Similarly to the approach for the Nikkei 225 index, we forward-fill all dates corresponding to American holidays where no trading takes place.

3.3 Summary Statistics

In the following section, we present summary statistics for all variables encompassed by the $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$ feature sets respectively for the entire sample period spanning January 1999 to August 2020. Further information and more detailed insights are provided in Appendix XXX.

4 Methodology and Empirical Design

4.1 Realized Variance and Quarticity

Let P_t denote the asset price. Following the realized-volatility literature, we view log-prices as evolving according to a continuous-time diffusion,

$$d\log(P_t) = \mu_t dt + \sigma_t dW_t, \quad (1)$$

where μ_t is the (possibly time-varying) drift, σ_t is the instantaneous volatility process, and W_t is a standard Brownian motion. The object of interest is the (latent) *integrated variance* over day t ,

$$IV_t = \int_{t-1}^t \sigma_s^2 ds, \quad (2)$$

which summarizes the cumulative variance generated by the continuous component of returns during the day.

Since IV_t is not directly observable, we estimate it using high-frequency intraday returns. Partition each trading day into M equally spaced intervals of length Δ (so that $M = 1/\Delta$) and define the intraday log-return as

$$r_{t,i} = \log(P_{t-1+i\Delta}) - \log(P_{t-1+(i-1)\Delta}), \quad i = 1, \dots, M. \quad (3)$$

The *realized variance* is then constructed as the sum of squared intraday returns,

$$RV_t = \sum_{i=1}^M r_{t,i}^2 = \underbrace{\sum_{i=1}^M (\log P_{t,t_i} - \log P_{t,t_{i-1}})^2}_{\text{Intraday variation (09:00–17:30)}} + \underbrace{(\log O_t - \log C_{t-1})^2}_{\text{Overnight variation}}. \quad (4)$$

Under standard regularity conditions and as $\Delta \rightarrow 0$, RV_t is a consistent estimator of IV_t .

In finite samples, however, realized variance is subject to non-trivial estimation error. In particular, the high-frequency limit theory implies the measurement-error representation

$$RV_t = IV_t + \eta_t, \quad \eta_t \mid IQ_t \sim \mathcal{N}(0, 2\Delta IQ_t), \quad (5)$$

where the error variance depends on the (latent) *integrated quarticity*

$$IQ_t = \int_{t-1}^t \sigma_s^4 ds. \quad (6)$$

Intuitively, IQ_t scales the uncertainty in RV_t : days with larger IQ_t imply a noisier estimate of IV_t .

Because IQ_t is unobserved, we approximate it with *realized quarticity*,

$$RQ_t = \frac{M}{3} \sum_{i=1}^M r_{t,i}^4, \quad (7)$$

which is consistent for IQ_t as the sampling frequency increases. Thus, RQ_t provides a feasible, time-varying proxy for the magnitude of the realized-variance measurement error and is central for distinguishing between days when RV_t is measured precisely versus imprecisely.

4.2 Benchmark Models

The benchmark models comprise of the HAR model, developed by (Corsi (2009)), and its extension; LogHAR (Corsi (2009)), LevHAR (Corsi and Renò (2012)), SHAR (Patton and Sheppard (2015)), and HARQ (Bollerslev et al. (2016)). Each model is described in more detail in the following subsections.

4.2.1 HAR Model

The Heterogeneous Autoregressive (HAR) model is a parsimonious time-series model designed to capture the long-term memory behavior of realized volatility using a simple linear regression framework. It was introduced by Corsi (2009) to model high-frequency based volatility measures such as realized variance. The model is motivated by the Heterogeneous Market Hypothesis, which argues that financial markets consist of agents operating at different investment horizons, such as short-term traders, medium-term portfolio managers, and long-term institutional investors.

Because these groups react to information over different time scales, volatility exhibits persistence across daily, weekly, and monthly horizons. The HAR model captures this multi-scale structure by explicitly including volatility components measured at different aggregation levels, thereby providing a simple approximation to long-memory dynamics.

$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-1}^{(w)} + \beta_m \overline{RV}_{t-1}^{(m)} + \varepsilon_t \quad (8)$$

Here, RV_{t-1} is simply yesterday's realized variance while $\overline{RV}_{t-1}^{(w)}$ and $\overline{RV}_{t-1}^{(m)}$ denote the lagged weekly and monthly averages of realized variance, respectively. These components are constructed as rolling means over the previous 5 and 22 trading days, thereby excluding non-trading days.

$$\overline{RV}_{t-1}^{(w)} = \frac{1}{5} \sum_{i=1}^5 RV_{t-i} \quad \text{and} \quad \overline{RV}_{t-1}^{(m)} = \frac{1}{22} \sum_{i=1}^{22} RV_{t-i} \quad (9)$$

The HAR model is often considered the baseline specification in the realized volatility literature, from which many extensions later have been developed.

4.2.2 LogHAR Model

The LogHAR model is simply a log-transformed version of the original HAR specification, estimated in logarithms rather than levels. The LogHAR model is typically applied because realized variance is strictly positive, highly right-skewed, and exhibits strong heteroskedasticity. Estimating the model in logarithms stabilizes the variance of the series, reducing the influence of extreme volatility spikes.

$$\log(RV_t) = \beta_0 + \beta_d \log(\overline{RV}_{t-1}) + \beta_w \log(\overline{RV}_{t-1}^{(w)}) + \beta_m \log(\overline{RV}_{t-1}^{(m)}) + \varepsilon_t \quad (10)$$

When the HAR model is estimated in logarithms, forecasts are produced for $\log(RV_t)$. However, because the exponential function is nonlinear, simply exponentiating the predicted value leads to a downward-biased forecast of RV_t . To correct for this, we apply the standard log-normal bias adjustment.

$$\widehat{RV}_t = \exp\left(\log(\widehat{RV}_t) + \frac{1}{2}\widehat{\sigma}^2\right) \quad (11)$$

4.2.3 LevHAR Model

The standard HAR model captures volatility across horizons. However, it ignores an important empirical fact that negative returns today increases tomorrow's volatility more than positive returns of the same magnitude. This is known as the leverage effect.

$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-1}^{(w)} + \beta_m \overline{RV}_{t-1}^{(m)} + \underbrace{\gamma_d r_{t-1}^- + \gamma_w \overline{r}_{t-1}^{(w)-} + \gamma_m \overline{r}_{t-1}^{(m)-}}_{\text{Leverage effect}} + \varepsilon_t \quad (12)$$

Here, $\gamma_d r_{t-1}^-$ denotes yesterday's daily negative return, defined as $\min(r_{t-1}, 0)$. This term captures the immediate impact of a negative shock on next-day volatility. Moreover, $\gamma_w \overline{r}_{t-1}^{(w)-}$ represents the average negative return over the previous 5 trading days corresponding to a week, capturing short-run cumulative downside pressure. Finally, $\gamma_m \overline{r}_{t-1}^{(m)-}$ denotes the average negative return over the past 22 trading days corresponding to a month, highlighting more persistent downward trends during periods when markets experience pro-longed financial distress.

4.2.4 SHAR Model

Closely related to the LevHAR model, the Semivariance HAR (SHAR) model introduced by [Patton and Sheppard \(2015\)](#) provides an alternative way of capturing asymmetry in volatility dynamics. While LevHAR incorporates negative returns directly to model the leverage effect, SHAR decomposes realized variance into its positive and negative

semivariance components, allowing volatility to respond differently to upside and downside variation.

$$RV_t = \beta_0 + \beta_d^+ RV_{t-1}^+ + \beta_d^- RV_{t-1}^- + \beta_w \overline{RV}_{t-1}^{(w)} + \beta_m \overline{RV}_{t-1}^{(m)} + \varepsilon_t \quad (13)$$

$$RV_{t-1} = RV_{t-1}^+ + RV_{t-1}^- \quad (14)$$

In this specification, RV_{t-1}^+ and RV_{t-1}^- denote the positive and negative realized semi-variances, constructed as the sum of intraday and overnight squared returns conditional on returns being positive and negative, respectively. This decomposition allows the model to distinguish between upside and downside volatility, reflecting the empirical finding that negative price movements typically exert stronger and more persistence effect on future volatility than positive movements of similar magnitude.

4.2.5 HARQ Model

The HARQ model was introduced by [Bollerslev et al. \(2016\)](#) and extends the standard HAR framework by explicitly accounting for the time-varying measurement error in realized variance. Since realized variance is a generated regressor, it contains estimation noise whose variance is proportional to integrated quarticity. Ignoring this induces an attenuation bias in the HAR coefficients. The HARQ model corrects this errors-in-variables problem by allowing the autoregressive coefficient on daily realized variance to vary with the square root of realized quarticity. When measurement error is small, persistence is higher; when measurement error is large, the influence of lagged realized variance is reduced. This adjustment leads to stronger average persistence and improved volatility forecasting.

$$RV_t = \beta_0 + \underbrace{\left(\beta_1 + \beta_{1Q} \sqrt{RQ_{t-1}} \right)}_{\beta_{1,t}} RV_{t-1} + \beta_2 RV_{t-1}^{(w)} + \beta_3 RV_{t-1}^{(m)} + \varepsilon_t. \quad (15)$$

Here, RQ_{t-1} represents yesterday's realized quarticity which makes the coefficient $\beta_{1,t}$ vary with time. In essence, $\sqrt{RQ_{t-1}}$ proxies the standard deviation of the measurement error in RV_{t-1} . As argued by [Bollerslev et al. \(2016\)](#), the HARQ adjustment is applied only to the daily lag because temporal aggregation in the weekly and monthly averages substantially reduces measurement error and, hence, attenuation bias, rendering additional Q-adjustments for those components largely unnecessary.

4.3 Regularized Linear Models

Regularized linear models are a set of extensions of the standard linear regression (OLS) models that incorporates penalty term to control model complexity and prevent overfitting. The general loss function for a regularized linear model can be written as:

$$\min_{\beta_0, \beta} \left\{ \sum_{t=1}^T (y_t - \beta_0 - X_t \beta)^2 + \lambda \cdot \phi(\beta) \right\} \quad (16)$$

In the setting above, the first component of the expression, the sum of squared residuals $\sum_{t=1}^T (y_t - \beta_0 - X_t \beta)^2$, corresponds to the standard ordinary least squares loss function and measures how well the model fits the observed data. In the context of this paper, y_t denotes the realized variance at time t , while X_t represents the vector of predictors, such as lagged realized variance and additional explanatory variables included in the extended feature set $\mathcal{M}_{ALL}$. Minimizing this first term of the equation would yield the OLS estimator, which is optimal in-sample but prone to overfitting when the number of predictors is large or when multicollinearity is present, both of prevalent in high-frequency volatility forecasting.

To address this issue, the model incorporates a penalty term $\lambda \cdot \phi(\beta)$, where $\phi(\beta)$ is a function of the regression coefficients and $\lambda \geq 0$ is a tuning parameter controlling the strength of regularization. Note that if $\lambda = 0$, the regularized linear model is the same as the standard linear regression model. This term thus imposes a cost on large coefficients, effectively shrinking them towards zero. The inclusion of this penalty introduces bias into the estimate but reduces their variance, with the objective to improve out-of-sample performance.

Different versions of $\phi(\beta)$ give rise to different regularization models. For instance, a quadratic penalty term corresponds to ridge regression (RR), an absolute value penalty terms leads to the LASSO (LA) regression, and a combination of both RR and LA yields the elastic net (EN). In the following subsections, we follow the approach of [Christensen et al. \(2023\)](#) and present these three regularized linear regression models.

4.3.1 Ridge Regression

The Ridge Regression (RR), originally crafted by [Hoerl and Kennard \(1970\)](#), aims to reduce variance of the model by adding a quadratic penalty term to shrinks all coefficients continuously toward zero, but never exactly to zero. This is known as the L_2 penalty term and thus penalizes large coefficients more heavily. The penalty term is detailed in equation 17.

$$\phi(\beta) = \sum_{j=1}^p \beta_j^2 \quad (17)$$

Adopting this technique can prove to be highly useful in improving volatility forecasting with feature sets where variables are highly correlated and the model risks overfitting noise.

4.3.2 LASSO Regression

The LASSO (LA) regression is another form of regularized linear model introduced by Tibshirani (1996). Unlike the ridge regression, the LA regression instead adopts what is called the L_1 penalty term to the original ordinary least squares regression, which sums the absolute value of the coefficients.

$$\phi(\beta) = \sum_{j=1}^p |\beta_j| \quad (18)$$

With this alternative set-up for the penalty term presented in equation 18, the loss function of the LA regression can effectively shrink coefficients to zero, effectively performing variable selection. This characteristic of the LA regression will prove efficient to identify the most relevant drivers of next-day volatility.

4.3.3 Elastic Net

The Elastic Net (EN) regression extends the regularization techniques presented in equation 17 and 18 by combining the L_1 and L_2 penalty terms in the loss function.

4.4 Tree-based Models

4.4.1 Bagging

4.4.2 Random Forest

4.4.3 Gradient Boosting

Gradient Boosting (GB) is a tree-based machine learning algorithm originally proposed Friedman (2001). Rather than fitting a single complex model, GB constructs an ensemble of shallow decision trees sequentially, where each tree is trained to correct the errors of the preceding ensemble. This iterative refinement procedure is grounded in the principle of functional gradient descent, where the negative gradient of the loss function with respect to the current prediction serves as the target for each new tree.

Formally, the GB algorithm is initialized with a constant prediction as described by the equation below.

$$\hat{f}^{(0)}(Z_t) = \arg \min_{\beta_0} \sum_{t \in \text{train}} (RV_t - \beta_0)^2, \quad (19)$$

Under the mean squared error loss function, the model reduces the sample mean of realized variance. At each subsequent iteration $b = 1, \dots, B$, a shallow regression tree is fitted to the residuals of the current ensemble, and the prediction is updated according to equation (20).

$$\hat{f}^{(b)}(Z_t) = \hat{f}^{(b-1)}(Z_t) + \nu \sum_{j=1}^{J_b} \hat{\gamma}_{jb} \mathbf{1}_{\{Z_t \in R_{jb}\}}, \quad (20)$$

Here, R_{jb} denotes the j -th terminal node of the b -th tree, $\hat{\gamma}_{jb}$ is the optimal step size with that node, and $\nu \in (0, 1]$ is a learning rate that controls the contribution of each tree to the final prediction. The final forecast is provided by the accumulated ensemble $\hat{f}(Z_t) \equiv \hat{f}^{(B)}(Z_t)$.

Each individual tree in the ensemble constitutes a so-called weak learner, a model only marginally better than a naive benchmark, but their sequential combination produces powerful nonparametric forecasting model capable of capturing nonlinearities and interaction effects in the predictor space without requiring these to be specified by the researcher *a priori*.

A well-known limitation of GB in financial applications is its sensitivity to outliers. Because the algorithm places disproportionate weight on large residuals when fitting each successive tree, extreme observation, such as volatility spikes during market stress episodes, can distort the learned function. This stands in contrast to Random Forests, which mitigates this issue through independent bootstrap sampling. Given this pronounced right-skewness and heavy tails of realized variance, this sensitivity is a relevant consideration in our setting, as noted by [Christensen et al. \(2023\)](#).

Following [Christensen et al. \(2023\)](#), the GB model is estimated using a rolling window scheme, where each forecast origin the hyperparameters are re-selected on the validation portion of the current window. Specifically, the tree depth is tuned over $\{1, 2\}$, the learning rate over $\{0.01, 0.1\}$, and the number of trees over $\{50, 100, \dots, 500\}$, with all remaining parameters set to their default values as proposed in the original implementation. The combination of minimizing the mean squared error on the validation set is retained, and the model is re-fitted on the full training window prior to generating the one-day-ahead forecast. Any negative forecast is replaced by zero, following standard practice in the realized variance literature.

4.5 Neural Network

Artificial neural networks (NNs) represent a class of machine learning algorithms inspired by the structure of the human brain. Unlike traditional linear models such as the HAR and its extensions, NNs are capable of approximating highly nonlinear and complex functional relationships between predictors and the response variable. This flexibility makes them particularly well-suited for financial time series, where the underlying data-generating process is known to be nonlinear, exhibits low signal-to-noise ratios, and is subject to structural breaks and regime changes over time. As noted by [Christensen et al. \(2023\)](#), the ability of NNs to capture nonlinearities in the dynamic structure of realized variance is a key source of forecast improvement relative to the linear HAR benchmark

models.

4.5.1 Feed-forward Neural Architecture

We proceed by implementing a feed-forward neural network, also known as a multilayer perceptron, in which information flows in a single direction from the input layer through one or more hidden layers to the output layer. Formally, the general equation for the l -th hidden layer is given by

$$a_t^{(\theta_{l+1}, b_{l+1})} = g_l \left(\sum_{j=1}^{N_l} \theta_j^{(l)} a_t^{(\theta_l, b_l)} + b^{(l)} \right), \quad 1 \leq l \leq L \quad (21)$$

In this equation, g_l is the activation function, $\theta^{(l)}$ is the weight matrix, $b^{(l)}$ is the bias vector serving as an activation threshold, N_l denotes the number of neurons in layer l , and $a_t^{(\theta_{l+1}, b_{l+1})}$ is the output of layer $l + 1$. The final forecast is constructed by composing the transformation across all layers.

$$\hat{f}(Z_t) = (g_L \circ \dots \circ g_1)(Z_t) \quad (22)$$

In accordance with [Christensen et al. \(2023\)](#), we use the Leaky Rectified Linear Unit (L-ReLU) of [Maas et al. \(2013\)](#) as the activation function for all hidden layers.

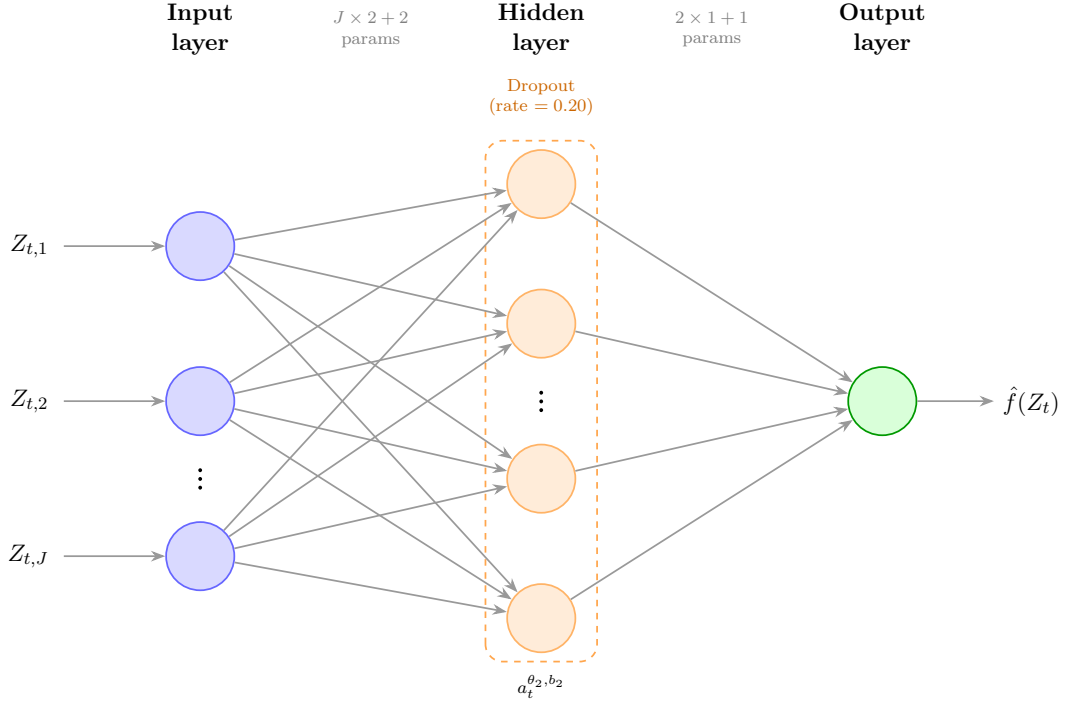
$$\text{L-ReLU}(x) = \begin{cases} cx & \text{if } x < 0 \\ x & \text{otherwise} \end{cases} \quad (23)$$

Here, we set $c = 0.01$ following the approach of [Christensen et al. \(2023\)](#). The L-ReLU is preferred over standard ReLU because it allows gradient-based learning even when the activation is negative, avoiding the so-called dying ReLU problem where neurons permanently output zero and cease to contribute to learning. The output layer uses a linear activation function, appropriate for a regression problem where the target variable is continuous.

Following the geometric pyramid structure of [Christensen et al. \(2023\)](#), we implement the NN1 architecture, visualized in Figure (4), as the simplest member of the neural network family. Despite its parsimonious structure, [Christensen et al. \(2023\)](#) demonstrate that even this shallow architecture consistently outperforms the HAR benchmark across all forecast horizons in their paper, suggesting forecasting gains are primarily driven by the network’s ability to capture nonlinearities in the dynamic structure of realized variance rather than by model depth per se. We therefore restrict our analysis to the NN1 architecture and do not implement the deeper variants considered in [Christensen et al. \(2023\)](#), namely NN2, NN3, and NN4.

Moreover, this decision is also motivated by computational feasibility. While [Christensen et al. \(2023\)](#) employ a fixed estimation for all network architectures, we extend the

Figure 4: NN1 Feed-Forward Neural Network Architecture



Notes: The figure illustrates the NN1 feed-forward neural network architecture following Christensen et al. (2023). The network receives J input features $Z_{t,1}, \dots, Z_{t,J}$, where $J = 3$ for the $\mathcal{M}_{\text{HAR}}$ feature set and $J = 20$ for the extended $\mathcal{M}_{\text{ALL}}$ feature set. The single hidden layer applies a Leaky ReLU activation function with negative slope $c = 0.01$ and is subject to dropout regularization with a keep rate of 0.8. The output layer returns the one-step-ahead forecast $\hat{f}(Z_t)$ via a linear activation function. We restrict our analysis to NN1 and do not consider deeper architectures (NN2, NN3, NN4), as Christensen et al. (2023) show that even this shallow network outperforms the HAR benchmark, and deeper architectures are computationally infeasible given our monthly rolling window re-estimation scheme.

paper by adopting a monthly rolling window re-estimation scheme, re-estimating the ensemble every 22 trading days to allow the network to adapt to new volatility regimes over time. This extension, while improving the comparability of the NN1 with the daily rolling HAR benchmark, increases computational costs proportionally with the number of model parameters, rendering deeper architectures computationally infeasible in our single-index setting. The NN1 architecture therefore represents the most appropriate balance between model flexibility and computational feasibility for our application.

5 Empirical Results

In the following sections, we present the empirical findings of our study. We begin by presenting the performance of each model using only the predictors available in the $\mathcal{M}_{\text{HAR}}$ feature set. We then proceed to compare these results with the results obtained from allowing the more advanced ML models to exploit nonlinearities and more complex patterns from the additional variables available in the extensive $\mathcal{M}_{\text{HAR}}$ feature set.

5.1 Results from $\mathcal{M}_{\text{HAR}}$ Feature Set

The following Table (1) illustrates the pairwise relative MSE performance of each model pair. We conduct the Diebold-Maraiona test to observe whether the difference in model performances is statistically significant or not.

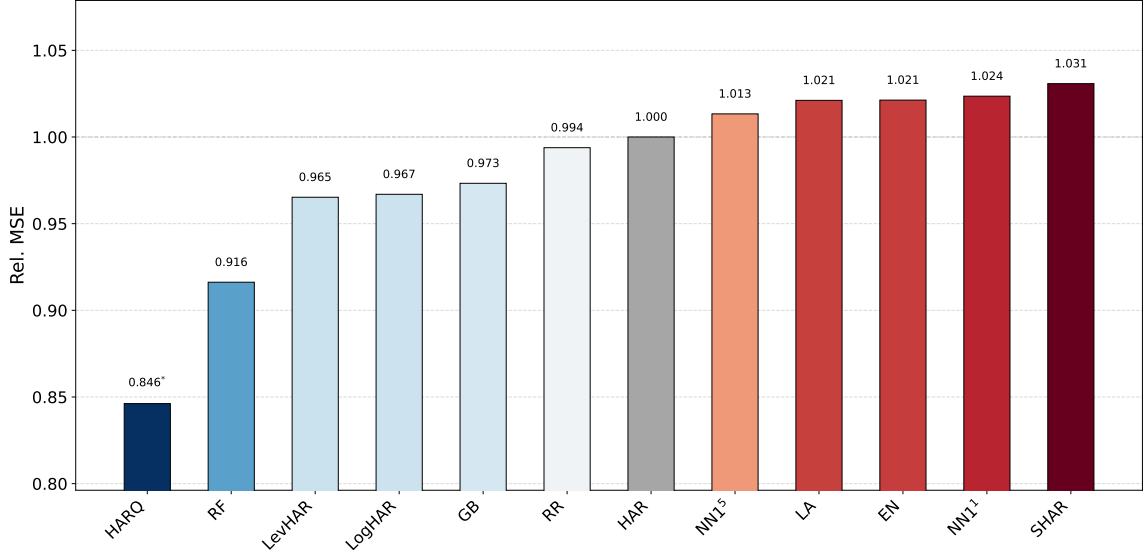
Table 1: Pairwise Relative MSE and Diebold-Mariano Tests — MHAR Feature Set

	Benchmark Models					Regularized Models			Tree-Based		Neural Networks	
	HAR	LogHAR	LevHAR	SHAR	HARQ	RR	LA	EN	RF	GB	NN1 ¹	NN1 ⁵
HAR	—	1.034	1.036	0.970	1.182*	1.006	0.979	0.979	1.091	1.027	0.977	0.987
LogHAR	0.967	—	1.002	0.938	1.143	0.973	0.947	0.947	1.055	0.994	0.945	0.954
LevHAR	0.965	0.998	—	0.936	1.141*	0.971	0.945	0.945	1.054	0.992	0.943	0.953
SHAR	1.031	1.066	1.068	—	1.218*	1.037	1.009	1.009	1.125	1.059	1.007	1.017
HARQ	0.846	0.875	0.877	0.821	—	0.851	0.829	0.829	0.924	0.869	0.827	0.835
RR	0.994	1.028	1.030	0.964	1.175*	—	0.973	0.973	1.085	1.021	0.971	0.981
LA	1.021*	1.056	1.058*	0.991	1.207**	1.028***	—	1.000	1.115*	1.049	0.998	1.008
EN	1.021	1.056	1.058*	0.991	1.207**	1.028***	1.000	—	1.115*	1.049	0.998	1.008
RF	0.916	0.948	0.949	0.889	1.083*	0.922	0.897	0.897	—	0.941	0.895	0.904
GB	0.973	1.007	1.008	0.944	1.150**	0.979	0.953	0.953	1.062**	—	0.951	0.961
NN1¹	1.024	1.059**	1.060	0.993	1.210*	1.030	1.002	1.002	1.117	1.052	—	1.010
NN1⁵	1.013	1.048*	1.050	0.983	1.198*	1.020	0.992	0.992	1.106	1.041	0.990	—

Notes: Each entry (i, j) reports the out-of-sample MSE of row model i relative to column model j , computed over the test period 2016-04-21 to 2020-08-06 (1,099 observations). A value below unity indicates that the column model j outperforms the row model i . Horizontal lines separate model categories. Significance stars denote rejection of equal predictive accuracy in favour of the column model based on one-sided Diebold-Mariano tests with Newey-West standard errors (5 lags). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

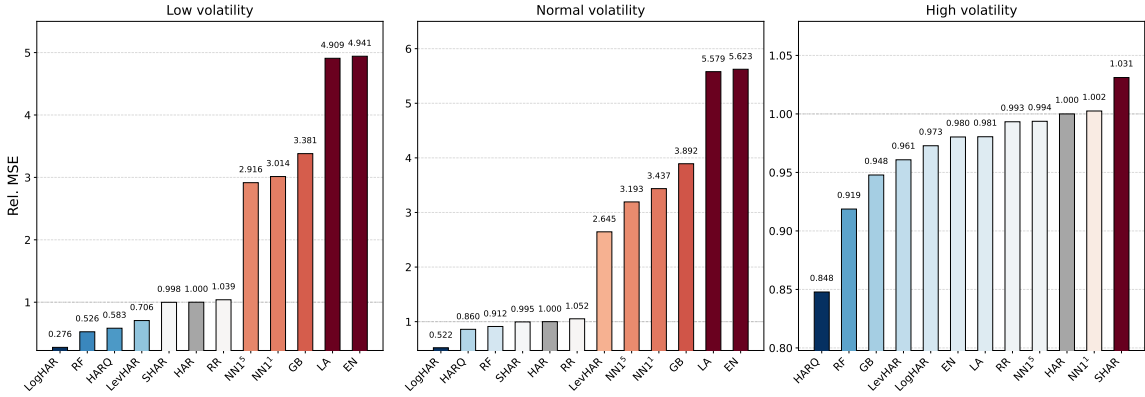
For illustrational purpose, we also present relative MSE performance with respect to the standard HAR model in Figure (5) for the entire out-of-sample period spanning 2016-04-21 to 2020-08-06. In addition, we decompose forecast accuracy across the realized variance distribution by splitting the test set into terciles based on observed realized variance levels. This reveals whether model outperformance is concentrated in calm markets, normal conditions, or stress episodes. Given that our test period spans both the low-volatility regime of 2017 and the Covid-19 spike of early 020, this decomposition is particularly informative..

Figure 5: Relative MSE of Competing Models under the $\mathcal{M}_{\text{HAR}}$ Feature Set



Notes: The figure reports out-of-sample relative mean squared error (Rel. MSE) for each model under the MHAR feature set, scaled relative to the HAR benchmark. A value below unity indicates that the model outperforms HAR, while values above unity indicate inferior performance. The dashed horizontal line marks the benchmark value of one. Significance stars denote one-sided Diebold–Mariano tests.

Figure 6: Relative MSE of Competing Models across Volatility Regimes under the $\mathcal{M}_{\text{HAR}}$ Feature Set



Notes: The figure reports out-of-sample relative mean squared error (Rel. MSE) across low-, normal-, and high-volatility regimes under the MHAR feature set, scaled relative to the HAR benchmark within each regime. The volatility regimes are defined using terciles of realized variance over the test sample. A value below unity indicates that the model outperforms HAR within the corresponding regime, whereas values above unity indicate inferior performance. The dashed horizontal line marks the benchmark value of one. Models are ordered within each panel according to their relative performance, facilitating comparison of ranking changes across volatility regimes.

5.2 Results from $\mathcal{M}_{\text{ALL}}$ Feature Set

6 Discussion

7 Concluding Remarks

8 Appendix

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